

## **FEATURE #1: SATELLITE-VIDEO FUSION AGENT**

This agent combines two different types of visual data to detect environmental anomalies. It uses satellite imagery from Google Earth Engine to scan large areas like forests and agricultural land. When it detects something unusual like a sudden change in vegetation or water levels, it immediately pulls video footage from drones or public CCTV cameras to confirm what it saw. The magic happens when CLIP embeddings convert both images and videos into the same mathematical space so they can be compared directly. The agent outputs real-time alerts with confidence scores and automatically captures coordinates and timestamps. Example: Satellite detects forest loss in Amazon. Agent fetches drone video of the location. Both confirm deforestation. Alert generated immediately.

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## **FEATURE #2: POINT CLOUD 3D ANALYSIS AGENT**

This agent processes LiDAR data which is like a three-dimensional photograph of a forest or landscape. Each point in the cloud represents a precise location in space with X, Y, and Z coordinates. The agent uses clustering algorithms to group points that belong to the same tree or building. It separates ground points from vegetation points. It measures distances between points to calculate tree heights accurately. It estimates biomass by analyzing the volume and density of vegetation. Unlike 2D satellite images that only show what's on top, point clouds reveal the entire vertical structure. Example: LiDAR scans a rainforest. Agent identifies 5000 individual trees, measures each one, groups them by height and density, calculates total carbon stored in biomass.

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## **FEATURE #3: HYPERSPECTRAL CARBON DETECTION AGENT**

This agent uses satellite bands that human eyes cannot see. Instead of just red, green, and blue, it accesses infrared, near-infrared, and short-wave infrared bands. It calculates indices like NDVI which reveals how healthy vegetation is by comparing near-infrared reflectance to red reflectance. It calculates NDWI to detect water content in soil and plants. It analyzes SWIR bands to measure stress indicators. Using spectral analysis, it estimates soil organic carbon by looking at the reflective properties that correlate with carbon content. The agent creates heatmaps showing which areas have high carbon stocks and which are degrading. Example: Satellite shows NDVI of 0.8 in pristine forest, 0.3 in degraded forest. Agent calculates soil carbon of 200 tons per hectare in healthy area, 20 tons in degraded area.

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#### **FEATURE #4: DISTRIBUTED IOT SENSOR NETWORK AGENT**

This agent coordinates 50-100+ IoT sensors deployed across a region. Each sensor measures temperature, humidity, air quality, soil moisture, or other environmental parameters. Instead of sending all data to a cloud server, each sensor runs anomaly detection locally using statistical methods. When a sensor detects something unusual, it alerts nearby sensors. The sensors form a mesh network where if one sensor fails, data still flows through the network via other sensors. The agent uses collective intelligence where one sensor's anomaly combined with nearby sensors' data creates a more reliable alert. Example: Soil moisture sensor in plot A detects dryness. Sensor B confirms. Sensor C shows temperature spike. Combined, agents predict drought 3 days before official weather service.

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#### **FEATURE #5: ORCHESTRATOR (MASTER COORDINATOR)**

This is the brain that runs all other agents simultaneously without them interfering with each other. It receives alerts from all agents and fuses them together. If satellite detects anomaly and sensor confirms it and point cloud measures it, orchestrator combines all three pieces of evidence before alerting users. It prioritizes which alerts matter most. It ensures when one agent is slow, others keep running. It maintains an audit trail so you can see exactly what each agent detected and when. It handles failures gracefully so if one agent crashes, others keep working. Example: At 10:00 AM, satellite detects forest change. At 10:05 AM, sensor network detects fire signature. At 10:10 AM, point cloud confirms tree loss. Orchestrator combines all three at 10:10 AM and sends alert with 99% confidence instead of three separate uncertain alerts.

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#### **FEATURE #6: WILDFIRE RISK DETECTION AGENT**

This agent predicts where wildfires will happen before they start. It uses thermal satellite data (MODIS) to detect temperature anomalies in forests. It calculates vegetation dryness using spectral indices. It integrates weather data like wind speed, humidity, and recent rainfall. It identifies forests that burned historically to know which areas are at highest risk. It combines all this data into a risk score from 0-100. When risk exceeds threshold, it alerts fire departments and sends evacuation warnings. It predicts not just whether fire will happen, but also the direction it will spread based on wind patterns. Example: Australian forest shows NDVI dropping 50%, recent rainfall zero, wind speed 40 km/h, historically burned 10 years ago. Agent calculates wildfire risk of 92/100 and predicts spread toward populated area.

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## **FEATURE #7: FLOOD RISK PREDICTION AGENT**

This agent predicts flooding by analyzing multiple data sources. It uses SAR satellite (Sentinel-1) which penetrates clouds and rain to detect water bodies and their changes. It analyzes terrain elevation to understand water flow paths. It monitors rainfall predictions and current precipitation. It correlates historical flood zones with current water levels. It calculates risk of flooding for each pixel in a region. When river levels rise above threshold, it alerts downstream communities. It predicts water arrival time and peak flood height. Example: River rises 2 meters, satellite shows water spreading into agricultural areas, rainfall forecast shows 50mm more rain coming, elevation analysis shows 5000 people downstream. Agent predicts flood peak in 6 hours, affecting 5000 people, water depth 3 meters.

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## **FEATURE #8: AIR QUALITY REAL-TIME MONITOR AGENT**

This agent aggregates air quality data from hundreds of sensors and satellite measurements. It calculates Air Quality Index (AQI) by measuring PM2.5 (finest particles), PM10 (larger particles), NO2 (nitrogen dioxide), O3 (ozone), SO2 (sulfur dioxide). It identifies pollution sources by analyzing where high concentrations originate. It predicts hour-by-hour how air quality will change based on wind patterns and meteorological models. It calculates health impact by estimating how many people will be affected by poor air quality. It sends personalized alerts to vulnerable populations like children and elderly. Example: City shows PM2.5 of 250 (hazardous), NO2 elevated from traffic, wind pattern shows pollution will hit hospital area in 2 hours. Agent alerts hospital to prepare and notifies asthmatics in the area.

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## **FEATURE #9: WATER QUALITY ASSESSMENT AGENT**

This agent monitors rivers and lakes by analyzing satellite water color and in-situ measurements. It detects chlorophyll concentrations which indicate algal blooms that can be toxic. It measures turbidity which shows suspended sediment and pollution. It infers water pH by analyzing light absorption at different wavelengths. It identifies pollution sources by analyzing where bad water originates and flows. It calculates water quality index combining multiple parameters. It predicts whether water is safe for drinking, fishing, or recreation. It tracks long-term water health trends. Example: Lake shows high chlorophyll and low clarity, suggesting algal bloom. Water quality agent

measures toxin levels, identifies agricultural runoff as source, recommends fishing ban, predicts bloom will peak in 3 days.

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#### **FEATURE #10: DEFORESTATION REAL-TIME DETECTOR AGENT**

This agent continuously monitors forests worldwide using satellite imagery. It compares images from different dates to detect tree loss. It distinguishes between natural tree loss (logging, fire) and permanent conversion to other land use. It identifies illegal logging by detecting sudden loss in protected areas. It quantifies how many trees were lost and how much carbon was released. It uses multi-temporal data to understand change patterns. It provides immediate alerts with coordinates of deforestation events. It maintains historical records showing deforestation trends year over year. Example: Amazon satellite shows 500 hectares of forest lost in protected area in past 24 hours. No fire detected. Deforestation agent alerts IBAMA, calculates 50,000 tons of carbon released, provides GPS coordinates, identifies as likely illegal logging.

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#### **FEATURE #11: REAL-TIME DISASTER RESPONSE ORCHESTRATOR AGENT**

This agent detects multiple types of disasters simultaneously: earthquakes from seismic data, floods from satellite and rainfall, wildfires from thermal data, tsunamis from seismic + buoy data. When disaster occurs, it immediately assesses damage using satellite imagery from before and after. It calculates how many people are affected by analyzing population distribution. It optimizes evacuation routes using graph algorithms considering blocked roads. It allocates emergency resources like hospitals, fire stations, rescue teams to highest-need areas. It continuously updates as new information arrives. It coordinates with emergency services. Example: Earthquake magnitude 7.2 strikes. Within 2 minutes, agent detects it, identifies 10,000 buildings with likely damage, finds 50,000 people in affected area, recommends evacuation routes, identifies 15 hospitals still functional, assigns rescue teams to highest-damage zones.

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#### **FEATURE #12: OCEAN PLASTIC TRACKING AGENT**

This agent detects plastic patches floating in oceans using satellite imagery. It analyzes water color and texture to identify plastic accumulation zones. It integrates ocean current models to predict where plastic will drift. It predicts accumulation zones weeks in advance. It prioritizes cleanup operations by identifying largest patches and most

accessible locations. It tracks plastic movement over time to understand patterns. It estimates total plastic mass in detected patches. It coordinates with cleanup organizations to deploy collection vessels. Example: Satellite detects 10,000 ton plastic patch in Pacific. Current models show it will hit Australia coast in 2 weeks. Agent alerts cleanup organizations, calculates fastest collection route, estimates 5-day window for collection before it disperses, coordinates vessel deployment.

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### **FEATURE #13: METHANE EMISSION HOTSPOT MAPPER AGENT**

This agent detects methane plumes from space using hyperspectral satellite data. It identifies point sources like oil/gas facilities, coal mines, landfills by analyzing where methane concentration peaks. It estimates emission rates by analyzing concentration gradients around sources. It flags facilities with unexpectedly high emissions suggesting leaks. It monitors whether facilities are complying with regulations. It correlates facility characteristics with emission levels to identify problematic operations. It sends alerts to environmental regulators and facility operators. Example: Satellite shows methane plume over oil refinery with concentration 50 ppb vs background 2 ppb. Agent estimates 500 tons methane per day leaked, flags as priority for inspection, notifies EPA, calculates this facility has twice the emissions of similar facilities.

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### **FEATURE #14: CARBON SEQUESTRATION CALCULATOR AGENT**

This agent calculates how much carbon is stored in forests and vegetation. It uses LiDAR data to measure tree volumes. It uses spectral data to identify tree species and health status. It applies scientific formulas called allometric equations that convert tree volume to biomass. It converts biomass to carbon content using standard conversion factors. It calculates net annual carbon sequestration by comparing measurements over time. It estimates carbon credit value using market prices. It validates calculations against ground-truth tree inventory measurements. It provides certification-ready reports for carbon markets. Example: Forest area of 1000 hectares has 50,000 trees averaging 50 cubic meters volume. Agent calculates 25,000 tons biomass, 12,500 tons carbon stored, 500 tons sequestered annually, worth \$7.5 million in carbon credits at \$15/ton.

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## **FEATURE #15: ESG/CLIMATE DASHBOARD AND REPORTING AGENT**

This agent aggregates outputs from all 14 other agents into a single unified dashboard. It displays real-time alerts, metrics, trends. It generates automated reports in CSRD and TCFD formats for corporate compliance. It creates carbon accounting statements showing scope 1, 2, 3 emissions. It visualizes environmental metrics on maps and time series charts. It provides stakeholder reports that non-technical people can understand. It tracks progress toward environmental targets. It enables data export in multiple formats for further analysis. It maintains audit trails for regulatory compliance.

Example: Company executive logs into dashboard. Sees carbon footprint is 5000 tons annually, down 10% from last year. Views map showing top three emission sources. Reads automatically generated CSRD report showing compliance status, downloads carbon accounting spreadsheet for auditors, sees 80% progress toward net-zero target.